

CySA+ Analyst Simulation Student Guide

Log Correlation: Phish → Compromise → Beacons

What this activity is

This simulation is designed to teach you **how a SOC analyst thinks**, not to test memorization. All data in this lab is **instructor-created training content** and reflects **real analyst workflows**, not certification exam questions.

You are practicing:

- Log correlation
- Timeline analysis
- Identifying malicious patterns
- Choosing appropriate response actions

You are **not expected to know everything yet**.

How to Approach This Lab (Read First)

If you feel stuck, slow down and follow this order:

1. Start with **Email logs**
2. Move to **Proxy logs (GET first, then POST)**
3. Look for **repeated outbound patterns**
4. Confirm with **Firewall logs**
5. Then answer the questions

Do not jump between tabs randomly. Analysts follow evidence step by step.

Step 1: Email Gateway Log (Initial Delivery)

Go to the **Email Gateway Log** tab.

Look for:

- Emails marked as **delivered**
- Same sender, subject, and URL sent to multiple users

What this tells you:

- A phishing attempt was sent
- This establishes the **starting point of the incident**
- Not everyone who received the email clicked it

You will not answer questions from this tab directly. This log provides **context**, not conclusions.

Step 2: Proxy Log – Identify Who Clicked (GET Requests)

Switch to the **Web Proxy Log** tab.

Turn on:

- METHOD: GET

Look for:

- GET requests to the **same external domain**
- Multiple users accessing the same suspicious URL

How to think about it:

- GET = someone **visited** a page
- Each unique user here counts as a **click**

This helps answer:

- How many users clicked the phishing link?

Tip:

- Count **unique users**, not total rows.

Step 3: Proxy Log – Identify Possible Compromise (POST Requests)

Turn off GET and turn on:

- METHOD: POST

Look for:

- POST requests to the same external domain
- POST activity that occurs **after** a GET from the same user

Why this matters:

- POST often means **credentials submitted or payload staged**
- Not every click results in compromise

This helps determine:

- Which users may be compromised

Step 4: Identify Beaconing (Ongoing Communication)

Still in the Proxy Log, look for **patterns over time**.

What beaconing looks like:

- Repeated outbound traffic
- Small, consistent payload sizes
- Regular or near-regular intervals
- Same external host or domain
- Often the same User-Agent

Helpful actions:

- Sort by timestamp
- Follow one user at a time
- Enable “Suspected Beaconing” if available (Practice Mode only)

Important:

- Some repetitive traffic is **intentional noise** to throw you off
- Updates and telemetry can look regular but are legitimate
- Focus on **sequence**, not just repetition

Step 5: Firewall / Egress Log – Confirm Destination IP

Switch to the **Firewall / Egress Log** tab.

Look for:

- Outbound connections from internal IPs
- Repeated connections to the same external IP
- Consistent ports (commonly 80 or 443)

Why this matters:

- Confirms **actual outbound communication**
- Helps identify the **C2 destination IP**
- Corroborates what you saw in the proxy logs

Step 6: Identify Patient Zero

Patient Zero is:

- The **earliest user** that shows signs of compromise

How to identify:

- Sort proxy logs by timestamp
- Look for the first user who transitions from normal browsing into POST or beaconing behavior

Patient Zero is not always:

- The first person who received the email
- The first person who clicked

Step 7: Kill Chain Stage (Concept Check)

You are not memorizing definitions here.

Think simply:

- Repeated communication back to attacker infrastructure = **Command and Control (C2)**

This occurs **after compromise**, not during delivery.

Step 8: Select the Best Next Action

Ask yourself:

- Is the threat ongoing?
- Is there risk of continued damage?
- Do we need to preserve evidence?

Good analyst actions usually:

- Contain the threat
- Prevent further communication
- Preserve evidence for investigation

Avoid:

- Ignoring confirmed malicious patterns
- Rebooting or patching before containment
- Actions that destroy evidence

If You Get Stuck

Try these in order:

1. Filter logs by one method at a time
2. Sort by timestamp
3. Follow one user from beginning to end
4. Use the Hint button (Practice Mode)

This lab is about **learning how to think**, not speed.

Final Reminder

If you can explain **why** you chose an answer using evidence from the logs, you are doing this correctly.

Mistakes are expected. Reasoning is the real skill you are building.